DATA ANALYTICS PROJECT PROPOSAL

# The Anatomy of a Modern Breakup: What Reddit Tells Us About Love Gone Wrong

A Multi-Dataset Sentiment & Engagement Analytics Pipeline

Academic Year 2025 – 2026 | Data Analytics Course
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## 1. Team Members & Roles

| Name | Primary Responsibility |
|---|---|
| Muhammad Sharjeel Zahid | Data Preprocessing, Feature Engineering & Streamlit Dashboard |
| Okah Ahone Ebwekoh | Exploratory Data Analysis (EDA) & Problem Formulation |
| Tharun Kumar Reddy V.B. | Sentiment Scoring, Modelling & Evaluation |

## 2. Dataset Summary

This project combines two publicly available text datasets from Kaggle, each sourced from a different Reddit community, to provide complementary analytical perspectives:

- **Broad Dataset — "Unveiling Relationship Dynamics with Reddit"** (r/relationship_advice): Macro-level metadata on general partnership issues including post scores, comment counts, and timestamped headlines.
- **Specific Dataset — "Reddit Break-Up Stories Dataset (2023–2026)"** (r/breakups): A text-heavy emotional archive of first-person breakup narratives, richer in body text for sentiment analysis.

Both datasets will be concatenated with a *source* flag column to distinguish their origin. Key shared features used across the pipeline are summarised below:

| Feature | Type | Description |
|---|---|---|
| title | Text | Post headline / short summary |
| body | Text | Full narrative text of the post |
| score | Integer | Net upvote count — primary target variable |
| comms_num | Integer | Total comment interactions on the post |
| timestamp | DateTime | Date and time the post was published |

## 3. Analytical Problem

This project addresses two distinct but related analytical sub-problems using the combined Reddit dataset:

- **Sub-problem 1 — Pattern Discovery (Unsupervised):** Identify recurring themes and emotional clusters within breakup posts (e.g. trust issues, communication breakdowns, infidelity) using text frequency analysis and keyword co-occurrence.
- **Sub-problem 2 — Engagement Prediction (Supervised):** Determine whether measurable post attributes — body length, sentiment polarity score, and post timing — can predict whether a post receives high community engagement, defined as a score above the dataset median (binary classification: high vs. low).

Raw text data is inherently noisy and unstructured, requiring a systematic preprocessing pipeline before any analysis can be conducted reliably. Model performance will be evaluated using Accuracy, F1-score, and Precision/Recall on an 80/20 train/test split.

## 4. Project Goals

- **Identify Patterns:** Extract recurring thematic keywords and emotional tones from post titles and bodies using word frequency analysis and word clouds.
- **Sentiment Scoring:** Assign polarity scores (positive / neutral / negative) to each post using VADER, then analyse how sentiment distributions differ across engagement tiers.
- **Predict Engagement:** Build a classification model evaluating whether post length, sentiment score, and publish time can predict high user engagement (score above median).
- **Visualise Insights:** Develop an interactive Streamlit dashboard presenting word clouds, chronological sentiment trends, score distributions (PMF/CDF), and model performance metrics.

## 5. Tools & Technologies

| Category | Tools |
|---|---|
| Data Preparation | Python, Jupyter Notebooks, Pandas, NumPy |
| Storage & Querying | PostgreSQL (raw vs. processed text schemas for SQL-based filtering) |
| Text Processing | VADER (sentiment scoring), text cleaning with Pandas / re |
| Modelling | Scikit-learn — Logistic Regression / Decision Tree; Accuracy & F1 evaluation |
| Visualisation & UI | Streamlit, Matplotlib, Seaborn, WordCloud |

## 6. Expected Deliverables

- Cleaned and merged dataset files saved to data/processed/
- Feature-engineered dataset including sentiment scores, post length, and time features
- Jupyter Notebook documenting the full analytical pipeline — notebooks/main_analysis.ipynb
- Modular Python scripts for data cleaning and model evaluation
- Interactive Streamlit dashboard with word clouds, sentiment trends, and score distributions
- Final structured report (Final_Report.docx) following university formatting standards

## 7. Project Timeline

| Week | Focus Area | Key Tasks | Lead |
|---|---|---|---|
| Week 1 | Data Acquisition & Setup | Download both Kaggle datasets; concatenate with source flag; seed PostgreSQL database; initial inspection | Sharjeel |
| Week 2 | EDA & Preprocessing | Text cleaning (punctuation, stopwords, URLs); PMF/CDF of scores & comment counts; handle missing values | Okah |
| Week 3 | Feature Engineering & Modelling | VADER sentiment scoring; derive post-length & publish-hour features; binary label creation; train classifier; evaluate F1 & accuracy | Tharun |
| Week 4 | Dashboard & Report | Streamlit dashboard assembly; visualisation refinement; final report compilation and submission | All |

*Both datasets are publicly available on Kaggle and were collected from Reddit under their respective platform terms. All data used is anonymised aggregate text.*